

The MQ-Calculus:

A Calculus for Quantum Processes Involving Intermittent Measurement

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1 Introduction and Motivation

We introduce a calculus for processes that involve measurements interspersed throughout the computation. The key intuition is that *communication = measurement*. In the calculus presented below this is represented explicitly in the **COMM** rule.

2 Related Work

The calculus is related to the CQP calculus of Gay and Nagarajan [1]. The primary difference is that their calculus has an explicit measurement operator, whereas this calculus insists that measurement happens when a quantum state is communicated from one agent to another.

3 Technical Presentation

3.1 Syntax

$$\theta ::= [0, 2\pi)$$

States.

$$\begin{aligned} S, T ::= & |0\rangle \mid |1\rangle \mid v \\ & \mid H(S) \mid F(\theta)(S) \mid R(\theta)(S) \\ & \mid \text{CTRL-}F(\theta)(S, T) \mid \text{CNOT}(S, T) \mid \text{CTRL}(U)(S, S, S, \dots)(T, T, T, \dots) \\ & \mid S \otimes T \mid \mathbb{C} S \mid S + T \end{aligned}$$

Processes.

$$Q ::= \bigotimes_j x_j!(S) \quad (\text{output; when } j \text{ ranges over an empty domain this is } \mathbf{0})$$

$$P ::= \text{for} \left(\bigotimes_j m_j \leftarrow \bigotimes_j x_j \right) \{Proc\} \quad (\text{input; when } j \text{ ranges over an empty domain this is } \mathbf{0})$$

$$Proc ::= Q \mid P \mid \sum_i p_i Proc \mid Proc \mid Proc \quad \text{where } p_i \in \mathbb{R}^+$$

3.2 Hypothesis

$$p(P_1 \mid P_2) = p P_1 \mid p P_2$$

3.3 Distributing Sums over Parallel Composition

Note that we have

$$\begin{aligned} & \sum_{i_1} p_{1,i_1} \left(\bigotimes_j x_{i_1,j}!(S_{i_1}) \right) \mid \sum_{i_2} p_{2,i_2} \left(\mathbf{for} \left(\bigotimes_j m_{i_2,j} \leftarrow \bigotimes_j x_{i_2,j} \right) \{Proc_{i_2}\} \right) \\ &= \sum_{i_1, i_2} p_{1,i_1} p_{2,i_2} \left(\bigotimes_j x_{i_1,j}!(S_{i_1}) \right) \mid \left(\mathbf{for} \left(\bigotimes_j m_{i_2,j} \leftarrow \bigotimes_j x_{i_2,j} \right) \{Proc_{i_2}\} \right) \end{aligned}$$

This will be useful in writing the full comm rule.

3.4 The Comm Rule

We build the comm rule in steps. The driving intuition is that

$$p_1 \bigotimes_{j \in J} x_j!(S) \mid p_2 \mathbf{for} \left(\bigotimes_{j \in J} m_j \leftarrow \bigotimes_{j \in J} x_j \right) \{Proc\} \longrightarrow p_1 p_2 \sum_r |\langle r | S \rangle|^2 Proc\{r/m\}$$

where S is a $|J|$ -qubit state.

Thus the comm rule in full is:

Comm:

$$\frac{\sum_{i_1} p_{1,i_1} \left(\bigotimes_{j_1} x_{i_1,j_1}!(S_{i_1}) \right) \mid \sum_{i_2} p_{2,i_2} \left(\mathbf{for} \left(\bigotimes_{j_2} m_{i_2,j_2} \leftarrow \bigotimes_{j_2} x_{i_2,j_2} \right) \{Proc_{i_2}\} \right)}{\sum_{i_1, i_2} p_{1,i_1} p_{2,i_2} \sum_r |\langle r | S_{i_1} \rangle|^2 Proc_{i_2}\{r/m_{i_2}\}} \bigotimes_j x_{i_1,j} = \bigotimes_j x_{i_2,j}$$

where the outer sum ranges over all pairs (i_1, i_2) such that the channel tuples match, i.e. $\bigotimes_j x_{i_1,j} = \bigotimes_j x_{i_2,j}$, and the inner sum over r ranges over the $|J|$ -qubit computational basis states. For each such pair, the probability weight is the product $p_{1,i_1} p_{2,i_2}$, the state S_{i_1} is measured in the computational basis yielding outcome r with probability $|\langle r | S_{i_1} \rangle|^2$, and the continuation $Proc_{i_2}\{r/m_{i_2}\}$ is the body of the input with the measurement outcome substituted for the bound pattern variables.

4 Examples

1. Simple communication:

$$x!(S) \mid \mathbf{for}(s \leftarrow x)\{P\}$$

2. Iterated Hadamard:

$$x!(|0\rangle) \mid \mathbf{for}(s \leftarrow x) \left\{ x!(Hs) \mid \mathbf{for}(s \leftarrow x) \{x!(Hs)\} \right\}$$

3. Collapsed form:

$$x!(|0\rangle) \mid \mathbf{for}(s \leftarrow x)\{x!(H H s)\}$$

4. General circuit form:

$$wires!(|0 \cdots 0\rangle) \mid \mathbf{for}(initialState \leftarrow wires)\{output!(U_{\text{circuit}} initialState)\}$$

4.1 Notational Conventions

$$\begin{aligned} 1\ x!(0) \otimes 1\ y!(0) &= 1\ (xy)!(00) \\ (H \otimes I)(1\ x!(0) \otimes 1\ y!(0)) &= \frac{1}{\sqrt{2}}\ (xy)!(00) + \frac{1}{\sqrt{2}}\ (xy)!(10) \\ \text{CNOT}\left(\frac{1}{\sqrt{2}}\ (xy)!(00) + \frac{1}{\sqrt{2}}\ (xy)!(10)\right) &= \frac{1}{\sqrt{2}}\ (xy)!(00) + \frac{1}{\sqrt{2}}\ (xy)!(11) \\ \text{CNOT}\left((H \otimes I)(1\ x!(0) \otimes 1\ y!(0))\right) &\mid \mathbf{for}(v \leftarrow x)\{X\} \mid \mathbf{for}(w \leftarrow y)\{Y\} \end{aligned}$$

5 Analysis of Wigner's Friend Paradox

We model the Wigner's friend thought experiment in the MQ-calculus. Three agents—the cat, the friend, and Wigner—each hold one qubit.

Setting up the cat state.

$$cat!(H |0\rangle) \quad // \text{ sets up cat state}$$

Note that in the literature $H |0\rangle |0\rangle |0\rangle$ is also written $|+00\rangle$. Combining the three single-qubit outputs:

$$cat!(H |0\rangle) \otimes friend!(|0\rangle) \otimes wigner!(|0\rangle) = (cat \otimes friend \otimes wigner)!(H |0\rangle |0\rangle |0\rangle)$$

MWI (no collapse). Under a many-worlds interpretation, no measurement occurs and the unitary propagates through all three qubits:

$$(cat \otimes friend \otimes wigner)!(\text{CNOT}_{friend, wigner} \text{CNOT}_{cat, friend} H |0\rangle |0\rangle |0\rangle)$$

Copenhagen (sequential measurement). Under a Copenhagen interpretation, each communication step triggers a measurement. In the MQ-calculus this is expressed as nested input/output:

$$\begin{array}{l}
(cat \otimes friend \otimes wigner)!(H |0\rangle |0\rangle |0\rangle) \\
| \\
\mathbf{for}(catState, friendState \leftarrow cat \otimes friend) \{ \\
\quad friend!(CNOT_{cat, friend} |catState\rangle |friendState\rangle) \\
\quad | \\
\quad \mathbf{for}(friendState, wignerState \leftarrow friend \otimes wigner) \{ \\
\quad \quad wigner!(CNOT_{friend, wigner} |friendState\rangle |wignerState\rangle) \\
\quad \} \\
\}
\end{array}$$

The key distinction is that in the MWI version, the entire circuit is applied unitarily to the joint state before any output, while in the Copenhagen version, each **for**($\dots$) constitutes a measurement: the comm rule fires, the state collapses, and subsequent operations act on the post-measurement state. The MQ-calculus makes this difference syntactically explicit. That is, in the MQ-calculus there is no paradox. There are two different, but equally valid programs, and either may be executed.

6 Conclusions and Future Work

It's uncommon in communications like this to leave the principal motivation of the work to the conclusion, and yet because in this case it involves future work, this is exactly what we have done. In other papers Wells has argued there is a slight refactoring of the Leifer-Milner-Sewell constructions that gives a full featured account of bisimulation in terms of contexts. That is, given a rewrite system we can calculate a set of contexts that label transitions in a labelled transition systems where bisimulation is a congruence. Colloquially, contexts define the notion of experiment, and the bisimulation equivalence classes form the collection of entities that are experimentally indistinguishable.

In light of Wells' construction it is natural to ask whether there is a rewrite system in which experiments correspond to measurements. The mq-calculus is a first step towards answering this in the positive. However, Wells' construction does not involve the continuum, whereas the mq-calculus clearly does. More work needs to be done to fit the notion of context-labelled transitions to the notion of physical experiment.

References

- [1] Simon J. Gay and Rajagopal Nagarajan. Communicating quantum processes. In Jens Palsberg and Martín Abadi, editors, *Proceedings of the 32nd ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages, POPL 2005, Long Beach, California, USA, January 12-14, 2005*, pages 145–157. ACM, 2005.

7 Appendix: Alternative Design Options

7.1 Classical Processes Acting on Quantum States

A configuration consists of:

1. A number n of allocated wires.
2. A 2^n -dimensional state $|\psi\rangle$.

A *quantum process* acts on a configuration.

7.2 De Bruijn Indices

$$\begin{aligned}
 P, Q &::= \mathbf{0} \\
 &| P \mid Q \\
 &| \mathbf{new} P \\
 &| U[i_0, \dots, i_{k-1}] ; P \\
 &| \mathbf{out} i \\
 &| \mathbf{in} i \{P, Q\}
 \end{aligned}$$

Shift operation.

$$\begin{aligned}
 \text{shift}(c, \mathbf{new} P) &= \mathbf{new} \text{shift}(c + 1, P) \\
 \text{shift}(c, P \mid Q) &= \text{shift}(c, P) \mid \text{shift}(c, Q) \\
 \text{shift}(c, U[is] ; P) &= U[\text{shift}(c, is)] ; \text{shift}(c, P) \\
 \text{shift}(c, \mathbf{out} i) &= \mathbf{out} \text{shift}(c, i) \\
 \text{shift}(c, \mathbf{in} i \{P, Q\}) &= \mathbf{in} \text{shift}(c, i) \{\text{shift}(c, P), \text{shift}(c, Q)\}
 \end{aligned}$$

7.3 Denotational Semantics

These processes act linearly on the distribution:

$$\llbracket U[i_0, \dots, i_{k-1}] ; P \rrbracket(n, |\psi\rangle) = \llbracket P \rrbracket(n, U|\psi\rangle)$$

where $U|\psi\rangle$ is shorthand for acting on those qubits.

$$\llbracket \mathbf{new} P \rrbracket(n, |\psi\rangle) = \llbracket P \rrbracket(n + 1, |0\rangle \otimes |\psi\rangle)$$

Parallel composition (interleaving semantics).

$$\llbracket \prod_i P_i \rrbracket(n, |\psi\rangle) = \text{random permutation } \sigma \left(\bigcirc_{\sigma(i)} \llbracket P_{\sigma(i)} \rrbracket \right)(n, |\psi\rangle)$$

$$\llbracket P \rrbracket(n, |\psi\rangle) \otimes \llbracket Q \rrbracket(m, |\phi\rangle) \longrightarrow \llbracket P \mid \text{shift}(Q)^n \rrbracket(n + m, |\psi\rangle |\phi\rangle)$$

Synchronization is measurement.

$$\llbracket \mathbf{out} i \mid \mathbf{in} i \{P, Q\} \rrbracket(n, |\psi\rangle) = \Pr(i=|0\rangle) \llbracket P \rrbracket(n, |\psi\rangle_{0/i}) + \Pr(i=|1\rangle) \llbracket Q \rrbracket(n, |\psi\rangle_{1/i})$$

where $\{x/i\}$ means we collapse qubit i of ψ to $|x\rangle$.

7.4 Structural Congruence

The usual equations hold:

$$\begin{aligned}P \mid \mathbf{0} &= P \\P \mid Q &= Q \mid P \\(P \mid Q) \mid R &= P \mid (Q \mid R) \\\mathbf{new} \, P \mid Q &= \mathbf{new} \, (P \mid \text{shift}(1, Q))\end{aligned}$$